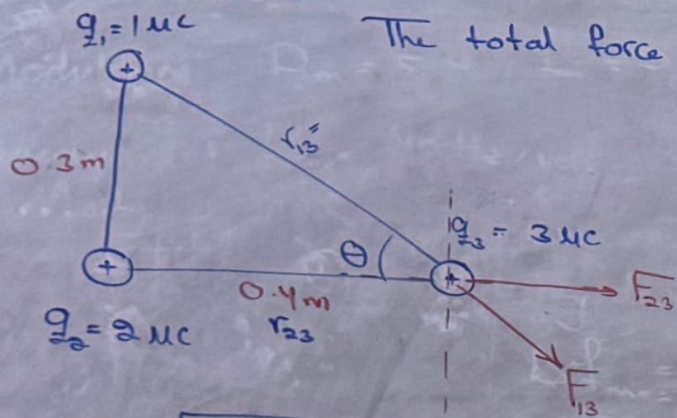


بسم الله الرحمن الرحيم

ex. 5



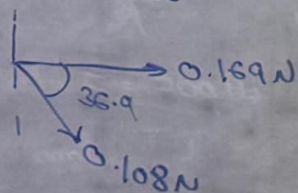
$$r_{12} = \sqrt{0.4^2 + 0.3^2} = 0.5 \text{ m}$$

$$\theta = \tan^{-1}\left(\frac{0.3}{0.4}\right) = 36.9^\circ$$

The total force on q_3 is:

$$F_{13} = k_e \frac{q_1 q_3}{r_{13}^2} = 9 \times 10^9 \times \frac{1 \times 10^{-6} \times 3 \times 10^{-6}}{0.5^2} = 0.108 \text{ N}$$

$$F_{23} = k_e \frac{q_2 q_3}{r_{23}^2} = 9 \times 10^9 \times \frac{2 \times 10^{-6} \times 3 \times 10^{-6}}{0.4^2} = 0.169 \text{ N}$$



$$F_x = 0.169 + 0.108 \cos 36.9 = 0.255 \text{ N}$$

$$F_y = -0.108 \sin 36.9 = -0.065 \text{ N}$$

$$F_R = \sqrt{0.255^2 + 0.065^2} = 0.263 \text{ N}$$

Electric field

7 protons and 3 electrons $\rightarrow q_{\text{net}} = 7e - 3e = 4e$

7 protons and 3 neutrons $\rightarrow q_{\text{net}} = 7e$

7 electrons and 3 protons $\rightarrow q_{\text{net}} = -7e + 3e = -4e$

$F_b = W_f = \cancel{P_f} \cancel{V_f} q$

↓
 V_o

$F_b = P_f \cancel{V_f} q$

بسم الحساب الثاني

Electric field

A region around a charge ()

The electric force divided by a charge

ratio N/C

$$E = \frac{F}{q}$$

$$E = k_e \frac{q}{r^2}$$

$$F = qE$$

$$F = ma$$

Electric field

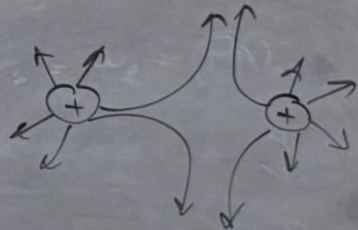
A region around a charge ()

The electric force divided by a charge

ratio N/C

$$E = \frac{F}{q}$$

$$E = k_e \frac{q}{r^2}$$



E lines begin from $+$
and terminate at $-$

$(F \cdot da)$ same as $\vec{E}$

$(F \cdot da)$ opposite as $\vec{E}$

E lines

- ① Positive: E lines are outward
Negative: E lines are inward

- ② No crossing

- ③ number of E lines $\rightarrow$ magnitude of charge
direction of E lines $\rightarrow$ sign of charge

